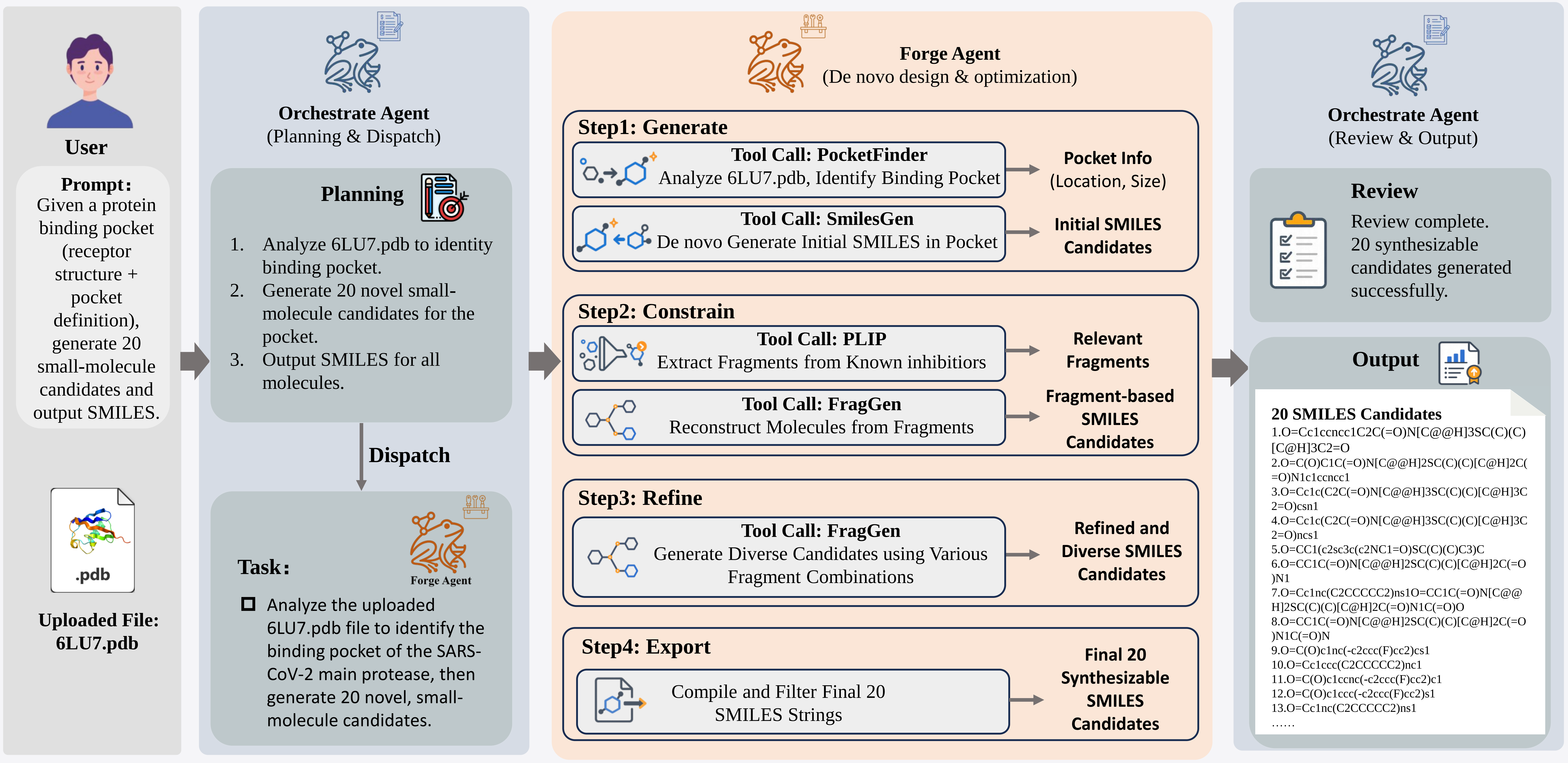

User
Prompt:
Given a protein binding pocket (receptor structure + pocket definition), generate 20 small-molecule candidates and output SMILES.
.pdb
Uploaded File:
6LU7.pdb
Orchestrate Agent
(Planning & Dispatch)
Planning
1. Analyze 6LU7.pdb to identity binding pocket.
2. Generate 20 novel small-molecule candidates for the pocket.
3. Output SMILES for all molecules.
Dispatch
Task:
Forge Agent
Analyze the uploaded 6LU7.pdb file to identify the binding pocket of the SARS-CoV-2 main protease, then generate 20 novel, small-molecule candidates.
Forge Agent
(De novo design & optimization)
Step1: Generate
Tool Call: PocketFinder
Analyze 6LU7.pdb, Identify Binding Pocket
Pocket Info
(Location, Size)
Tool Call: SmilesGen
De novo Generate Initial SMILES in Pocket
Initial SMILES Candidates
Step2: Constrain
Tool Call: PLIP
Extract Fragments from Known inhibitors
Relevant Fragments
Tool Call: FragGen
Reconstruct Molecules from Fragments
Fragment-based SMILES Candidates
Step3: Refine
Tool Call: FragGen
Generate Diverse Candidates using Various Fragment Combinations
Refined and Diverse SMILES Candidates
Step4: Export
Compile and Filter Final 20 SMILES Strings
Final 20 Synthesizable SMILES Candidates
Orchestrate Agent
(Review & Output)
Review
Review complete. 20 synthesizable candidates generated successfully.
Output
20 SMILES Candidates
1.O=Cc1ccncc1C2C(=O)N[C@@H]3SC(C)(C)[C@H]3C2=O
2.O=C(O)C1C(=O)N[C@@H]2SC(C)(C)[C@H]2C(=O)N1c1ccncc1
3.O=Cc1c(C2C(=O)N[C@@H]3SC(C)(C)[C@H]3C2=O)csn1
4.O=Cc1c(C2C(=O)N[C@@H]3SC(C)(C)[C@H]3C2=O)ncs1
5.O=CC1(c2sc3c(c2NC1=O)SC(C)(C)C3)C
6.O=CC1C(=O)N[C@@H]2SC(C)(C)[C@H]2C(=O)N1
7.O=Cc1nc(C2CCCCC2)ns1O=CC1C(=O)N[C@@H]2SC(C)(C)[C@H]2C(=O)N1C(=O)O
8.O=CC1C(=O)N[C@@H]2SC(C)(C)[C@H]2C(=O)N1C(=O)N
9.O=C(O)c1nc(-c2ccc(F)cc2)cs1
10.O=Cc1ccc(C2CCCCC2)nc1
11.O=C(O)c1ccnc(-c2ccc(F)cc2)c1
12.O=C(O)c1ccc(-c2ccc(F)cc2)s1
13.O=Cc1nc(C2CCCCC2)ns1
……